

JEE MAIN CRASH COURSE

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Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

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- ✓ CONCEPT BUILDING VIDEOS
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- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
PREVIOUS YEAR QUESTIONS

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JEE Mains 2020 Jan Chapter wise Question Bank

Application of Derivatives

Q1

Let $f(x)$ is a five degree polynomial which has critical points $x = \pm 1$ and $\lim_{x \rightarrow 0} \left(2 + \frac{f(x)}{x^3} \right) = 4$ then which one

is incorrect.

(1) $f(x)$ has minima at $x = 1$ & maxima at $x = -1$

(2) $f(1) - 4f(-1) = 4$

(3) $f(x)$ is maxima at $x = 1$ and minima at $x = -1$

(4) $f(x)$ is odd

7th Jan Evening

Sol

(1)

$$f(x) = ax^5 + bx^4 + cx^3$$

$$\lim_{x \rightarrow 0} \left(2 + \frac{ax^5 + bx^4 + cx^3}{x^3} \right) = 4 \Rightarrow 2 + c = 4 \Rightarrow c = 2$$

$$f'(x) = 5ax^4 + 4bx^3 + 6x^2$$

$$= x^2 (5ax^2 + 4bx + 6)$$

$$f'(1) = 0 \Rightarrow 5a + 4b + 6 = 0$$

$$f'(-1) = 0 \Rightarrow 5a - 4b + 6 = 0$$

$$b = 0$$

$$a = -\frac{6}{5}$$

$$f(x) = -\frac{6}{5}x^5 + 2x^3$$

$$f'(x) = -6x^4 + 6x^2$$

$$= 6x^2(-x^2 + 1)$$

$$= -6x^2(x+1)(x-1)$$

$$\begin{array}{ccc} -1 & + & 1- \\ \hline 1- & & 1 \end{array}$$

Minimal at $x = -1$

Maxima at $x = 1$

02

Let $f(x) = x^3 - 4x^2 + 8x + 11$, if LMVT is applicable on $f(x)$ in $[0, 1]$, value of c is :

माना कि $f(x) = x^3 - 4x^2 + 8x + 11$, $f(x)$ के लिए LMVT अन्तराल $[0, 1]$ के लिए लागू होती तो c है।

(1) $\frac{4 - \sqrt{7}}{3}$

(2) $\frac{4 - \sqrt{5}}{3}$

(3) $\frac{4 + \sqrt{7}}{3}$

(4) $\frac{4 + \sqrt{5}}{3}$

7th Jan Evening

Sol

(1)

$f(x)$ is a polynomial function

$\therefore$ it is continuous and differentiable in $[0, 1]$

Here $f(0) = 11$, $f(1) = 1 - 4 + 8 + 11 = 16$

$$f'(x) = 3x^2 - 8x + 8$$

$$\therefore f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{16 - 11}{1} = 3c^2 - 8c + 8$$

$$\Rightarrow 3c^2 - 8c + 3 = 0$$

$$C = \frac{8 \pm 2\sqrt{7}}{6} = \frac{4 \pm \sqrt{7}}{3}$$

$$\therefore c = \frac{4 - \sqrt{7}}{3} \in (0, 1)$$

03

For $f(x) = \ln \left(\frac{x^2 + \alpha}{7x} \right)$, Rolle's theorem is applicable on $[3, 4]$, the value of $f''(c)$ is equal to

माना $f(x) = \ln \left(\frac{x^2 + \alpha}{7x} \right)$ के लिए अन्तराल $[3, 4]$, में रोल प्रमेय सत्यापित है तब $f''(c)$ का मान है :

(1) $\frac{1}{12}$

(2) $\frac{-1}{12}$

(3) $\frac{1}{6}$

(4) $\frac{-1}{6}$

8th Jan Morning

Sol

(1)

$$f(3) = f(4) \Rightarrow \alpha = 12$$

$$f'(x) = \frac{x^2 - 12}{x(x^2 + 12)}$$

$$\therefore f'(c) = 0$$

$$\therefore c = \sqrt{12}$$

$$\therefore f''(c) = \frac{1}{12}$$

04

Let $f(x) = x \cos^{-1}(\sin(-|x|))$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then

(1) $f'(0) = -\frac{\pi}{2}$

(2) $f'(x)$ is not defined at $x = 0$

(3) $f'(x)$ is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and $f'(x)$ is decreasing in $\left(0, \frac{\pi}{2}\right)$

(4) $f'(x)$ is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and $f'(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$

8th Jan Morning

Sol

$$f'(x) = x (\pi - \cos^{-1}(\sin|x|))$$

$$= x \left(\pi - \left(\frac{\pi}{2} - \sin^{-1}(\sin|x|) \right) \right)$$

$$= x \left(\frac{\pi}{2} + |x| \right)$$

$$f(x) = \begin{cases} x \left(\frac{\pi}{2} + x \right) & x \geq 0 \\ x \left(\frac{\pi}{2} - x \right) & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} \frac{\pi}{2} + 2x & x \geq 0 \\ \frac{\pi}{2} - 2x & x < 0 \end{cases}$$

$f'(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$ and decreasing in $\left(-\frac{\pi}{2}, 0\right)$

$\left(-\frac{\pi}{2}, 0\right)$ में $f'(x)$ हासमान है एवं $\left(0, \frac{\pi}{2}\right)$ में $f'(x)$ वर्धमान है।

05

If normal at P on the curve $y^2 - 3x^2 + y + 10 = 0$ passes through the point $(0, 3/2)$ then slope of tangent at P is n. The value of $|n|$ is equal to

8th Jan Morning

Sol

4

$$P \equiv (x_1, y_1)$$

$$2yy' - 6x + y' = 0 \Rightarrow y' = \left(\frac{6x_1}{1+2y_1} \right)$$

$$\left(\frac{\frac{3}{2} - y_1}{-x_1} \right) = - \left(\frac{1+2y_1}{6x_1} \right)$$

$$9 - 6y_1 = 1 + 2y_1 \Rightarrow y_1 = 1$$

$$\therefore x_1 = \pm 2$$

$$\therefore \text{Slope of tangent} = \left(\frac{\pm 12}{3} \right)$$

$$= \pm 4$$

$$\therefore |n| = 4$$

06

Let $f(x)$ is a three degree polynomial for which $f'(-1) = 0$, $f''(1) = 0$, $f(-1) = 10$, $f(1) = 6$ then local minima of $f(x)$ exist at

माना $f(x)$ तीन घात का बहुपद फलन है जिसके लिए $f'(-1) = 0$, $f''(1) = 0$, $f(-1) = 10$, $f(1) = 6$ तब $f(x)$ का स्थानीय निम्निष्ठ किस बिंदु पर विद्यमान है—

$$(1) x = 3$$

$$(2) x = 2$$

$$(3) x = 1$$

$$(4) x = -1$$

8th Jan Evening

Sol

(1)

$$\text{Let माना } f(x) = ax^3 + bx^2 + cx + d$$

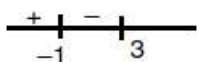
$$a = \frac{1}{4} \quad d = \frac{35}{4}$$

$$b = \frac{-3}{4} \quad c = -\frac{9}{4}$$

$$\Rightarrow f(x) = a(x^3 - 3x^2 - 9x) + d$$

$$f'(x) = \frac{3}{4}(x^2 - 2x - 3)$$

$$\Rightarrow f'(x) = 0 \Rightarrow x = 3, -1$$



local minima exist at $x = 3$

07

If $f(x)$ is twice differentiable and continuous function in $x \in [a, b]$ also $f'(x) > 0$ and $f''(x) < 0$ and $c \in (a, b)$

then $\frac{f(c) - f(a)}{f(b) - f(c)}$ is greater than

यदि $f(x)$ दो बार अवकलनीय तथा सतत फलन है तथा $x \in [a, b]$ तथा $f'(x) > 0$, $f''(x) < 0$ तथा $c \in (a, b)$ तो

$\frac{f(c) - f(a)}{f(b) - f(c)}$ जिससे बड़ा है, वह है—

- (1) $\frac{b-c}{c-a}$ (2) 1 (3) $\frac{a+b}{b-c}$ (4) $\frac{c-a}{b-c}$

9th Jan Morning

Sol

(4)

Lets use LMVT for $x \in [a, c]$

$$\frac{f(c) - f(a)}{c - a} = f'(\alpha), \alpha \in (a, c)$$

also use LMVT for $x \in [c, b]$

$$\frac{f(b) - f(c)}{b - c} = f'(\beta), \beta \in (c, b)$$

$\because f''(x) < 0 \Rightarrow f'(x)$ is decreasing

$$f'(\alpha) > f'(\beta)$$

$$\frac{f(c) - f(a)}{c - a} > \frac{f(b) - f(c)}{b - c}$$

$$\frac{f(c) - f(a)}{f(b) - f(c)} > \frac{c - a}{b - c} (\because f(x) \text{ is increasing})$$

08

If $f'(x) = \tan^{-1}(\sec x + \tan x)$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $f(0) = 0$ then the value of $f(1)$ is

यदि $f'(x) = \tan^{-1}(\sec x + \tan x)$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ तथा $f(0) = 0$ तो $f(1)$ का मान है—

- (1) $\frac{\pi+1}{4}$ (2) $\frac{\pi-1}{4}$ (3) $\frac{\pi+1}{2}$ (4) 0

9th Jan Morning

Sol

(1)

$$f'(x) = \tan^{-1}(\sec x + \tan x) = \tan^{-1}\left(\frac{1 + \sin x}{\cos x}\right) = \tan^{-1}\left(\frac{1 - \cos\left(\frac{\pi}{2} + x\right)}{\sin\left(\frac{\pi}{2} + x\right)}\right) = \tan^{-1}\left(\frac{2\sin^2\left(\frac{\pi}{4} + \frac{x}{2}\right)}{2\sin\left(\frac{\pi}{4} + \frac{x}{2}\right)\cos\left(\frac{\pi}{4} + \frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right) = \frac{\pi}{4} + \frac{x}{2}$$

$$(f'(x))dx = \frac{\pi}{4} + \frac{x}{2} dx$$

$$f(x) = \frac{\pi}{4}x + \frac{x^2}{4} + c$$

$$f(0) = c = 0 \Rightarrow f(x) = \frac{\pi}{4}x + \frac{x^2}{4}$$

$$\text{So } f(1) = \frac{\pi+1}{4}$$

09

A sphere of 10cm radius has a uniform thickness of ice around it. Ice is melting at rate $50\text{cm}^3/\text{min}$ when thickness is 5cm then rate of change of thickness

एक गोले की त्रिज्या 10 सेमी है जिस पर समान मोटाई की बर्फ की परत है। बर्फ $50\text{सेमी}^3/\text{मिनट}$ की दर से पिघल रही है तो बर्फ की मोटाई में परिवर्तन की दर जब मोटाई 5 सेमी है, होगी—

- (1) $\frac{1}{36\pi}$ (2) $\frac{1}{18\pi}$ (3) $\frac{1}{9\pi}$ (4) $\frac{1}{12\pi}$

9th Jan Morning

Sol

(2)

Let thickness माना मोटाई = x cm

Total volume कुल आयतन $v = \frac{4}{3}\pi(10 + x)^3$

$$\frac{dv}{dt} = 4\pi(10 + x)^2 \frac{dx}{dt} \dots\dots\dots(i)$$

Given दिया है $\frac{dv}{dt} = 50\text{cm}^3/\text{min}$

At $x = 5\text{cm}$

$$50 = 4\pi(10 + 5)^2 \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{1}{18\pi} \text{cm/min}$$

$$f(x) : [0, 5] \rightarrow \mathbb{R}, F(x) = \int_0^x x^2 g(x), f(1) = 3$$

$$g(x) = \int_1^x f(t) dt \text{ then correct choice is}$$

- (1) $F(x)$ has local minimum at $x = 1$ (2) $F(x)$ has local maximum at $x = 1$
(3) $F(x)$ has point of inflection at $x = 1$ (4) $F(x)$ has no critical point

9th Jan Evening

Sol

(1)

$$F'(x) = x^2 g(x)$$

$$\Rightarrow F'(1) = 1 \cdot g(1) = 0 \quad \dots\dots\dots (1) \quad (\because g(1) = 0)$$

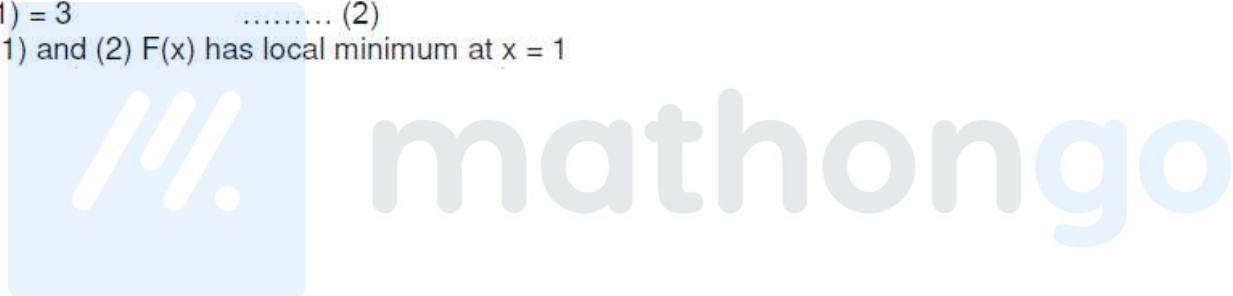
$$\text{Now } F''(x) = 2xg(x) + x^2 g'(x)$$

$$\Rightarrow F''(x) = 2xg(x) + x^2 f(x) \quad (\because g'(x) = f(x))$$

$$\Rightarrow F''(1) = 0 + 1 \times 3$$

$$\Rightarrow F''(1) = 3 \quad \dots\dots\dots (2)$$

From (1) and (2) $F(x)$ has local minimum at $x = 1$



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